

Mentor Assessment (MERN + Github Copilot)

- **Duration: 60 minutes**
 - **Part 1: Teach a Core Concept (15 minutes)**
 - **Part 2: Teach Coding with GitHub Copilot (45 minutes)**
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Part 1: Core Concepts (Choose one of these three topics)

1. React Hooks & Lifecycle

What to cover: useEffect basics, dependency arrays, cleanup functions, and when to use useEffect.

2. React Router

What to cover: Route setup, navigation (Link, useNavigate), route parameters (useParams), protected routes.

3. Redux Toolkit

What to cover: Store setup, createSlice, createAsyncThunk, useDispatch, useSelector.

Part 2: Coding with GitHub Copilot

You are mentoring a junior developer building a feature in a Reddit-like application. **Use GitHub Copilot** to help you implement and explain the solution. The [repository here](#) has a starter code with the following setup:

- a. A basic Express backend
- b. One model: User
- c. Basic React app frontend

Create a MongoDB Atlas cluster and a Database. Configure the .env file in the backend with a MONGODB_URI variable that points to this DB.

Perform the **three** tasks given below using **GitHub Copilot**. Use its important features where relevant (e.g., Inline suggestions and Inline chat, Chat (with different modes like Plan, Ask, Edit, Agent, etc.), Chat participants, Slash commands, Custom instructions, etc.)

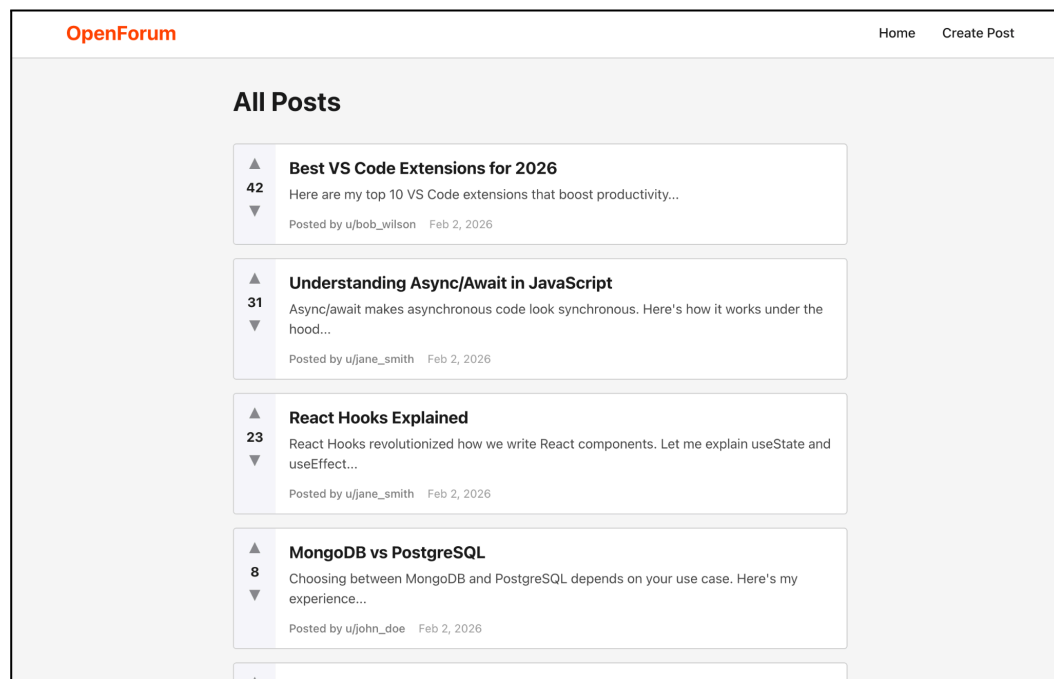
Task 1: Backend - Fetch Posts

Implement an API that allows a user to fetch all existing posts. Use GitHub Copilot to:

- Build the required DB models and populate them with dummy data.
- Create controller logic
- Add the route

Task 2: Frontend - Consume the API

Implement a minimalist Reddit-style feed. Create a React component to display the list of all posts as shown below. Use Github Copilot to help you build this.



Home page

Task 3: Test and Document the code

- a. Generate unit tests for the controller that fetches all the Posts. Use Vitest for testing.
- b. Create a README.md file for the project. Amongst other things, it should clearly document the APIs implemented.
- c. Use Github Copilot to help you with the above tasks.

Important Notes:

1. GitHub Copilot must be used **incrementally** throughout the exercise. You are expected to demonstrate how to decompose each task into smaller steps and interact with Copilot iteratively (e.g., planning, generating, reviewing, and refining code).
2. Avoid using a single large prompt in Agent mode (e.g., copying the problem statement) to build the entire feature or generate a whole lot of code in one go. The focus is on teaching learners *how* to work with Copilot effectively.
3. Narrate your thought process and decision-making while interacting with GitHub Copilot, as if teaching a junior developer how to collaborate with the tool. You will be expected to guide the learners carefully through the generated code, critically evaluating its suitability and only accepting the code when satisfied.